

AI Emotional Companion

Your Empathetic AI Friend, Always There
Anytime, without judgment

AI AGENTS

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The Problem - The Emotional Wellness Gap



- Rising stress, anxiety, loneliness rates: [WHO](#)
 - a. 1 in 5 adults in the U.S. experiences mental illness each year (NAMI, 2023).
- Stigma associated with seeking traditional help.
 - a. Over 50% of those affected do not receive treatment due to cost, stigma, or access issues (KFF, 2023).
- Cost and accessibility barriers to therapy/coaching.
 - a. Average wait time for a therapist: 3–6 weeks or more
 - b. \$100–\$200+ per session remains unaffordable for many, particularly Gen Z and lower-income groups (GoodRx Health)
- Lack of immediate, private, non-judgmental spaces for reflection
- **Relatable Scenario:** "Imagine feeling overwhelmed after a tough day at work or school, with no one immediately available to talk to..."

Our Solution: AI Emotional Companion

- Built **AI Emotional Companion** – a website (for now) designed to be that accessible, empathetic listener.
- Provides users with an AI-powered conversational partner, available instantly, anytime, anywhere
- **Core Value:**
 - a. Offer immediate, private, and non-judgmental emotional support
 - b. Help users feel heard and navigate their feelings.
- Not a replacement for therapy, but a readily available **tool for everyday emotional well-being.**

How It Works

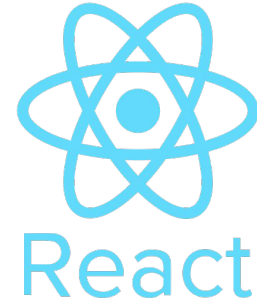
- During this hackathon, we focused on building the core experience



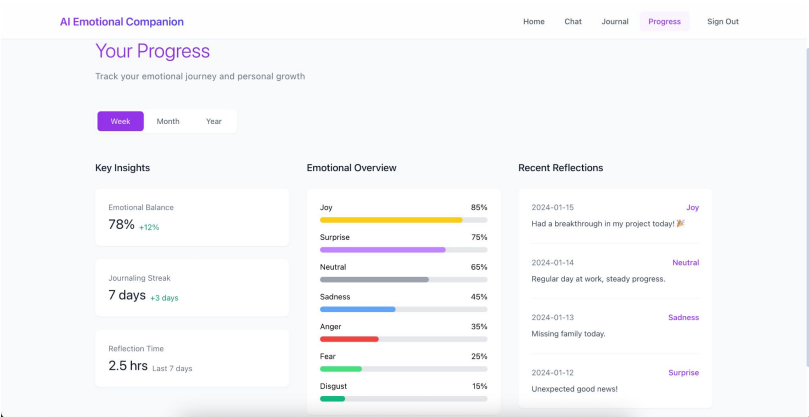
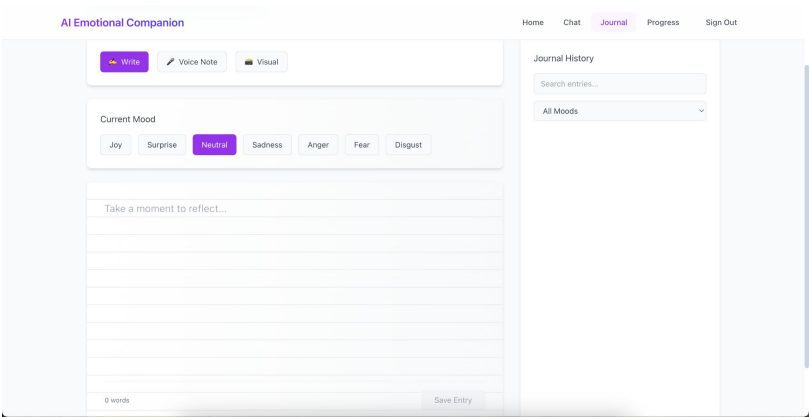
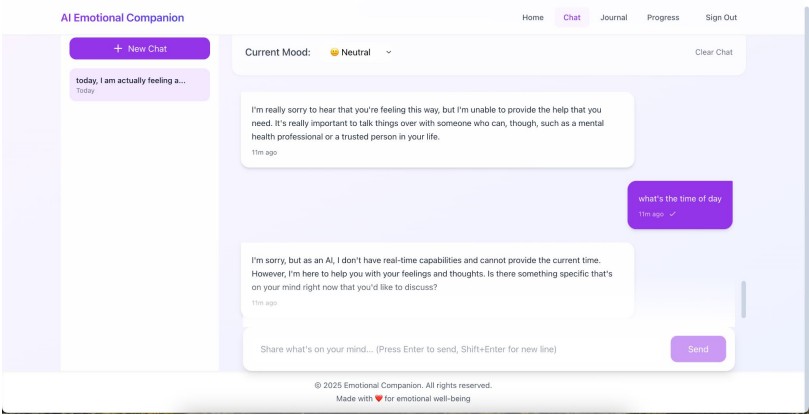
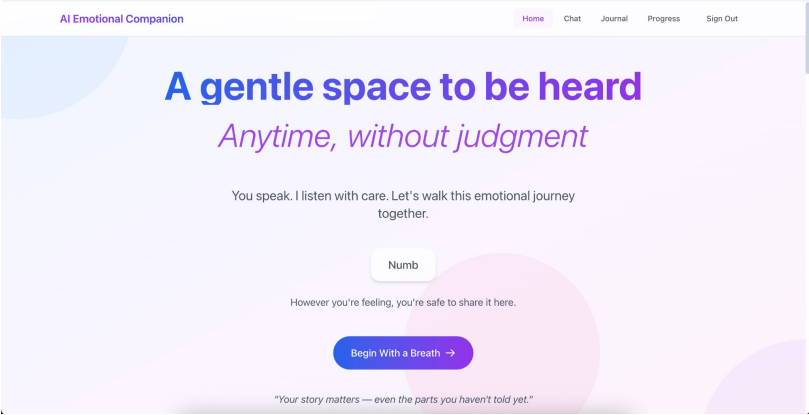
- Empathetic Chat Interface:** Users interact through a simple, intuitive chat.
- Voice Interaction:** For a more natural and hands-free experience, users can **speak**  **directly** with the AI (TO BE IMPLEMENTED)
- Integrated Journaling:** The app includes a dedicated journaling space.
- Video Entries:** Express yourself beyond words. Users can **record short video clips** directly in the app – perfect for capturing a moment or expressing complex feelings visually. (TO BE IMPLEMENTED)
- Image Sharing:** Sometimes a picture says it all. Users can **upload photos** from their gallery or take new ones to share experiences, environments, or visual representations of their mood. (TO BE IMPLEMENTED)
- Mood Logging:** Users can quickly log their mood, helping them track patterns over time
- Privacy Mention:** User conversations are treated with privacy in mind (TO BE IMPLEMENTED)

Technology Stack

- To bring AI Emotional Companion to life, we utilized:
 - a. **Frontend:** React
 - b. **Backend:** FastAPI
 - c. **Database:** MongoDB
 - d. **AI/Core Logic:** Emotional Analysis Model & OpenAI API



Live demo



Innovation

What makes AI Emotional Companion unique?

Current Model Capabilities

- **Focus on Empathy Quality:** Unlike basic chatbots, our focus is on nuanced empathetic responses.
- **Accessibility:** provides instant emotional first-aid, 24/7, lowering the barrier to seeking support
- **Simplicity:** Focused on a clean, uncluttered interface to make it easy and calming to use

Future Avenues for Integration:

- **Multi-Modal Support:** Allows users to express themselves through **text, voice, integrated journaling, video entries, AND image sharing.**
- **Privacy Focus-- Encryption:** Commitment to user privacy in a sensitive area like emotional well-being is a key differentiator

Model Architecture

How do we create a deep connection with the user?

- Have you ever asked a chatbot “Tell me about myself, psychoanalyze me based on what you know about me?”
- **We need a model architecture that is capable of extracting deep information from the user and storing it in its context window, BEYOND this capability.**
- We want the context window to go beyond superficial information about the user

Context Window

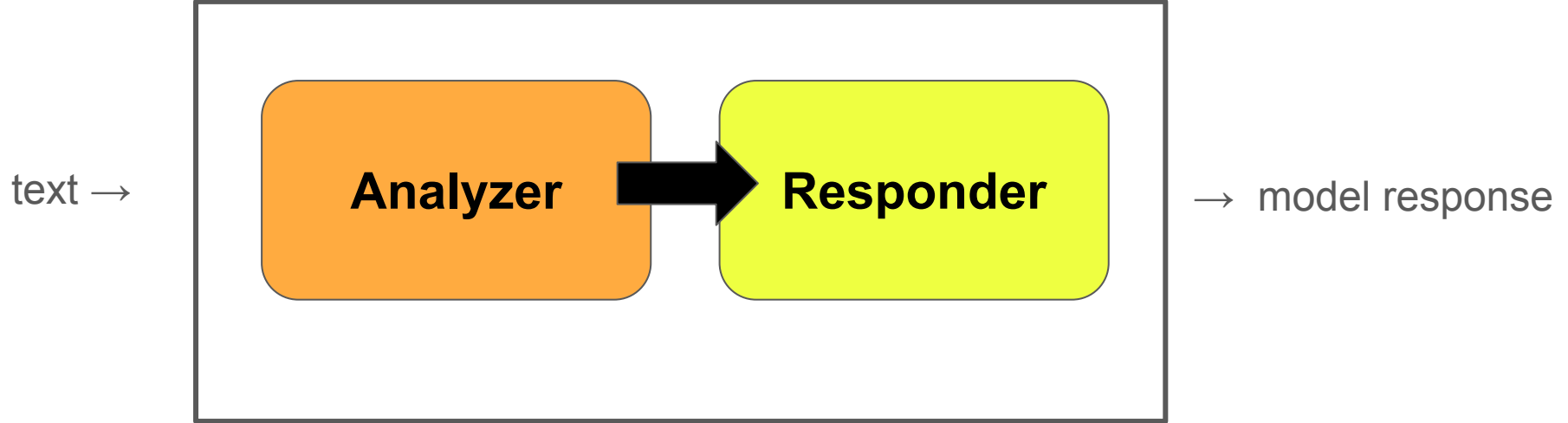
Traditional LLM Context Window

- ONLY the text, files
- LLM makes NO assumptions about the user, just reports and summarizes

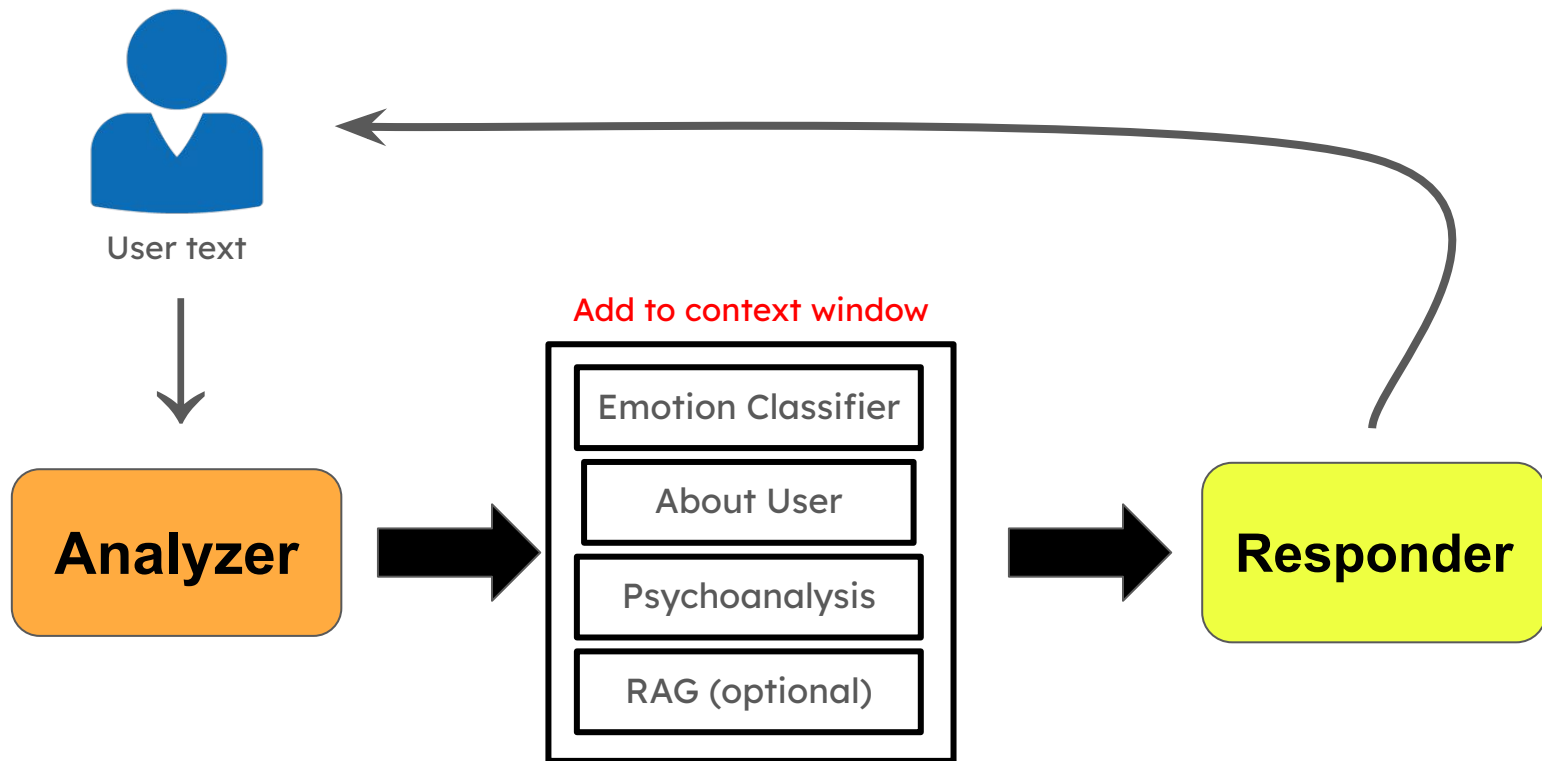
Emotional Companion AI

- Includes not just text between user and LLM but **MUCH** more...
 - Text sentiment (emotion data)
 - Psychoanalysis of user with hard and soft estimates
 - About user: conservative guesses about the user that aim to learn more about the user before generating an output
 - RAG Integration: (optional): draw from therapy textbooks or textbooks about best ways to deal with patients.

Primary Architecture

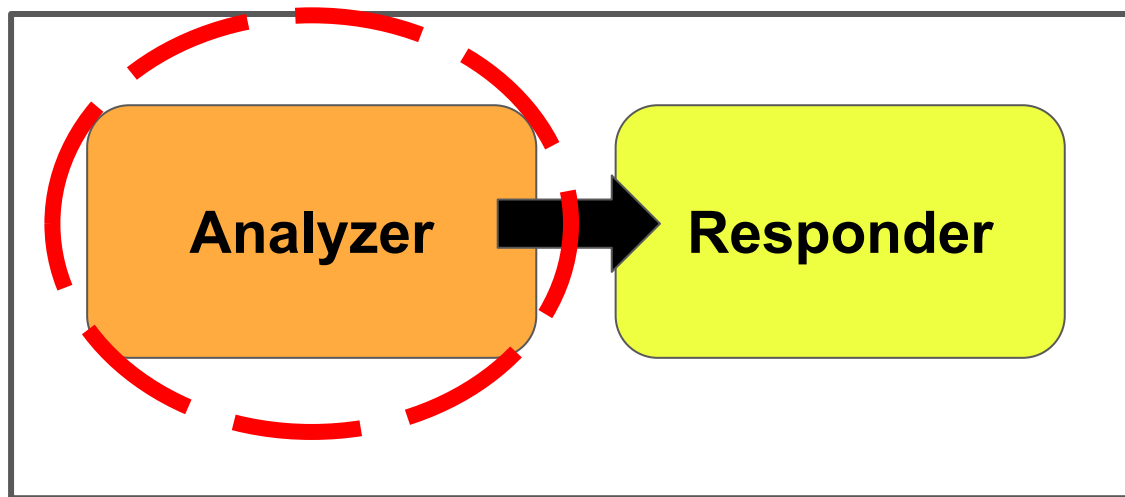


Overall pipeline



Analyzer

- Emotion Classifier
- About User (Certain, Uncertain [short term prob, long term prob])
- Psychoanalysis
- RAG (optional)



Analyzer: Emotion Sentiment Classifier

NRCLex (Top Emotions):

negative: 0.2631578947368421

sadness : 0.2631578947368421

VADER Sentiment (Polarity Scores):

neg : 0.20

neu : 0.65

pos : 0.15

compound: -0.52

Transformer-Based Emotion Classification (GoEmotions-like):

anger : 0.0062

disgust : 0.0779

fear : 0.0284

joy : 0.0032

neutral : 0.1246

sadness : 0.6979

surprise : 0.0617

***Transformer-based**

Happy – Clear Example

Today was amazing. I woke up feeling refreshed for once, and everything just flowed naturally. I got an email saying I'd been selected for a program I'd been dreaming about for months. My friends surprised me with a mini celebration, and we laughed until our cheeks hurt. I felt seen, supported, and genuinely alive — like everything I've been working for is finally coming together.

Happy – Nuanced Example

Sad – Clear Example

Sad – Nuanced Example

Analyzer: About User

Initial Form

Step 1 of 8 13%

What's your name, or how would you like me to address you?

Continue

Step 3 of 8 38%

How do you describe yourself, if you'd like to share?

Continue

About User ANALYSIS

```
about_user.txt
1  {
2    "certain": {
3      "The user values joy and friendship.": true,
4      "The user feels confident in their abilities.": true
5    },
6    "unsure": {
7      "The user may have recently formed new social connections.":
8        "short_term": 0.7,
9        "long_term": 0.5
10   },
11 }
```

The chatbot **stores certain and uncertain information about the user with respective probabilities**

- This will be used to give a more empathetic interaction between the model and user

Analyzer: Psychoanalysis (How does a therapist think?)

To form an intimate relationship with the user, the model **thinks deeply** about the user and adds its thoughts to the context window

- Long and short term probabilities included

```
    "cognitive_distortions": {  
      "Possible minimization of potential risks.": {  
        "short_term": 0.3,  
        "long_term": 0.2  
      }  
    },  
    "defense_mechanisms": {  
      "Might intellectualize emotions when excited.": {  
        "short_term": 0.4,  
        "long_term": 0.3  
      }  
    }  
  },  
  "maladaptive_patterns": {  
    "Might be prone to self-harm when stressed.": {  
      "short_term": 0.2,  
      "long_term": 0.1  
    }  
  }  
}
```



 1. Cognitive Distortions  2. Defense Mechanisms  3. Maladaptive Patterns  4. Inferred Beliefs / Self-Schema  5. Emotional Regulation Patterns

| good or bad thinking patterns - axioms that the user believes about the world or themselves. core beliefs -Cognitive Distortions -2. Defense Mechanisms - 3. Maladaptive Patterns - Inferred

Analyzer: RAG (optional, needs debugging)

- Based on the context window and the current prompt, the chatbot would **retrieve relevant information from therapy textbooks** to be the best emotional companion possible.

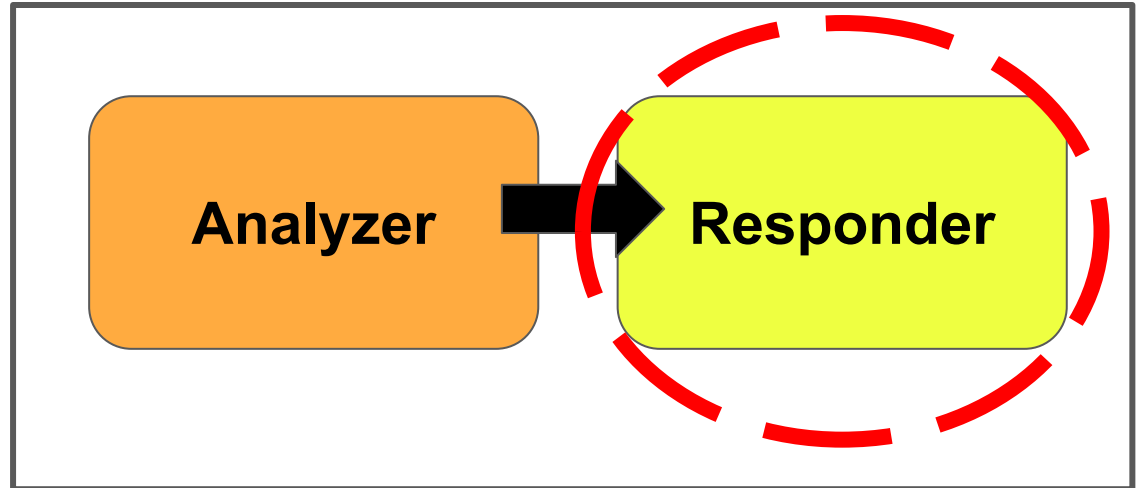
to be compliant and agreeable, at least on the surface. We may cheerfully go along with our partner's agenda, even when

was really awake. Gradually you began to turn into a sinister figure. You looked into my eyes with a light and said, "That's a lovely blue eye." I barely mumbled, "It's green," feeling that if you didn't know the color of my eyes it meant you didn't know me. I realized with a shock that I didn't know the color of your eyes, either. Brown, I thought, but I wasn't sure. Then you said to me, "What are the things I've told you?" I started to mumble, "Many things." You said, "No, I have told you nothing." I took this to mean that you are absolutely not responsible for anything I might do. These things made you seem sinister to me. You slowly began to change into another man who seemed to be a derelict, and I know I must get up. I struggled to awaken myself, and I finally succeeded. I ran to the door and ran out of the room, but there were a lot of people. In a mirror there I saw an utter ruin—I looked 80 years old and terribly ugly and I believe scarred. All the people were old and ugly. It was a village of discarded useless, and helpless people. A feeling of horror overcame me, and, as I stared at that face, I tried to comfort myself that it was only a nightmare and I would soon wake up, and I found it very difficult until I wasn't sure anymore if it was a nightmare or real. I finally woke up from the dream so frightened that I wanted to wake my husband, but I decided to try to calm down. I fell asleep again and had a second dream. I dreamed I had stayed up all night writing a paper you asked me to do. I started to bring it into the room you told me to. It was locked. I decided to have some coffee and came back. I did. This time your wife was in the room. She told me who she was. I said I knew. Then she told me she was your daughter's mother as though this made her a figure of great importance and dignity. This made me feel guilty and gave me the feeling that I could not see you anymore. She didn't want me to and in respect to her sacredness as a mother I couldn't. Had I not become alerted to the beginning transference, which certainly reflected an oedipal problem, I am convinced that my sinister qualities would have become so overwhelming in her unconscious mind that she would have discontinued therapy. As matters stood, we were able to engage in fruitful discussions to a satisfactory interpretation of her dream. "The task of interpreting a dream" says M. Hollander (1965), "is somewhat like understanding a poem." The therapist sensitizes himself or herself to the manner in which the dream is presented and attempts to divine why the patient is bringing up the material at

days afterward. [The patient is aware of his detachment, which, an important part of his character structure, seems to be a defense against further encroachment as well as a manifestation of hostility.] Th. Mm hmm. Pt. It's a very complex problem because, after all, people do like you for your qualities. They can't just like you really for nothing, and they can't like you for being a

Responder

- Responds based on all previous context
- Takes into consideration all aspects of user with nuanced context window
- Passes result to the frontend after running in the backend



Agent Criteria

Perceiving its environment

- Text
- Text sentiment
- About User
 - *Long and short term probabilities*
 - Enables realistic understanding of the patient: allows conservative guesses about patient
- Psychoanalysis
 - *Long and short term probabilities*
- RAG output (optional)

```
emotions.txt
1  {
2    "neutral": 0.5495,
3    "sadness": 0.1117,
4    "disgust": 0.104,
5    "surprise": 0.0788,
6    "anger": 0.0641,
7    "fear": 0.0514,
8    "joy": 0.0406
9  }
```

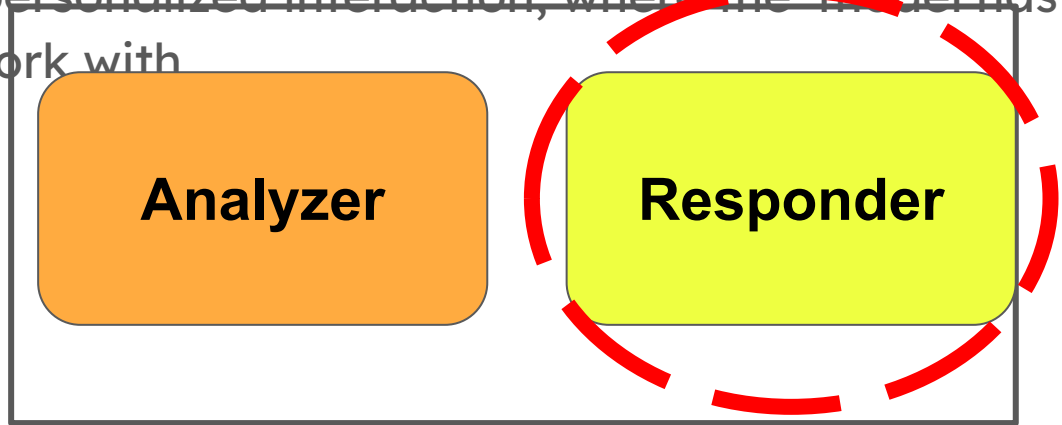
Reasoning or Making Decisions

- **Responder based on analyzer**
- About_User (form + analysis)
- Psychoanalysis
- Emotions (how to respond accordingly)
- Rag Output (optional)

```
psychoanalysis.txt
6   }
7   },
8   "axioms": {
9     "Believes effort leads to success.": {
10      "short_term": 0.8,
11      "long_term": 0.7
12    }
13  },
14  "cognitive_distortions": {
15    "Possible minimization of potential risks.": {
16      "short_term": 0.3,
17      "long_term": 0.2
18    }
19  },
20  "defense_mechanisms": {
21    "Might intellectualize emotions when excited.": {
22      "short_term": 0.4,
23      "long_term": 0.3
24    }
25  },
26  "maladaptive_patterns": {
27    "Mild dependency on external validation for emotional uplift.": {
28      "short_term": 0.35,
29      "long_term": 0.25
30    }
31  }
```

Executing Actions Part 1

- Responder-model based
- We ask the responder model to act as an emotional companion and we give it a very in-depth context window to act in accordance with the user
- This creates a far more personalized interaction, where the model has hard and soft probs to work with



Executing Actions Part 2

Extensions of our current work:

- Updating the visuals on the website with mood tracking
- Recommending next steps for the user, making use of Youtube API to recommend meditation videos/yoga, spotify API to recommend songs,

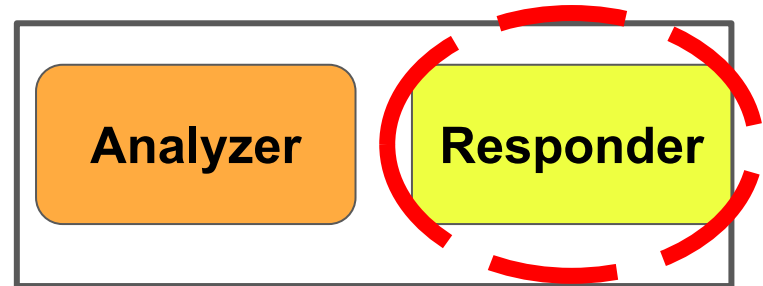


Demonstrating Autonomy

- Locally, we generate several text files to show the processes our model does in the background to enhance its own context window
 - emotions.txt
 - about_user.txt
 - psychoanalysis.txt
 - rag_output.txt
- On local demo, we can show these files generated in real-time, which will run throughout a user's conversation with the chatbot

Autonomy Extension

- Give the Responder more autonomy
- When you are trying to be emotionally available for a friend, it's often hard to determine whether your friend simply wants advice or whether they just want to be listened to, empathized with.
- **For even more autonomy, our group plans to integrate this decision-making process**
 - **The model will be able to decide for itself based on its psychoanalysis of the user if the user wants a solution or if they want to be empathized with.**



Hackathon Achievements

- Within the limited timeframe of this hackathon, our team successfully:
 - Built a functional prototype from concept to working demo.
 - Integrated the OpenAI API and engineered prompts for empathetic conversation flow.
 - Managed integration of multiple complex features (AI, Speech, Media Handling) on the backend/frontend.
 - Developed 3 core features: Chat, Mood Logging, journaling
 - Overcame challenges related to **frontend-backend integration and computational limitations**
 - Collaborated effectively across React & FastAPI & Emotional Analysis

Future Roadmap

- AI Emotional Companion has great potential. Our next steps would include:
 - **User Testing & Feedback:** Conducting thorough user testing to gather feedback and iterate on the design and functionality.
 - **Ethical Considerations:** Developing robust safety protocols, clear guidelines on the AI's limitations, and exploring partnerships with mental health professionals for oversight.
 - **Platform Expansion:** Integration with Mobile App.

The Team

- This project was brought to life by our dedicated team:



Conclusion

- Loneliness and the need for accessible emotional support are significant challenges.
- **AI Emotional Companion** offers a potential solution: an empathetic, private, and instantly available AI friend.
- We believe this tool can provide valuable support for everyday emotional well-being.

Code Material

- Emotional Model : <https://github.com/micahbaldonado/AI-Therapist>
- Front-end :
<https://gitlab.com/kristibaishya/ai-emotional-companion.git>
- Back-end :
<https://github.com/Kitiyaparnnn/Emotional-Companion-Back-end.git>